If m and n are integers such that $mn=1$, then either $m=1$ and $n=1$ or else $m=-1$ and $n=-1$. □

32) Show that these three statements are equivalent, where a and b are real numbers: i) a is less than b, ii) the average of a and b is greater than a, and iii) the average of a and b is less than b.

$\boxed{\text{Solution:}}$ Let $p_1 : a < b$ $\quad p_2 : \frac{a+b}{2} > a$ $\quad p_3 : \frac{a+b}{2} < b$

We will prove p_1, p_2 and p_3 are equivalent by showing that each of $p_1 \to p_2$, $p_2 \to p_3$ and $p_3 \to p_1$ are ~~[illegible]~~ True.

We will prove $p_1 \to p_2$ is True by a direct proof.

$\boxed{\text{Case 1:}}$ Let p_1 be True. $\therefore a < b \Rightarrow a + a < a + b \Rightarrow \frac{a+a}{2} < \frac{a+b}{2} \Rightarrow \frac{2a}{2} < \frac{a+b}{2}$

$\Rightarrow \boxed{a < \frac{a+b}{2}}$. $\boxed{\therefore p_2 \text{ is True}}$. By defn. of conditional, $\boxed{p_1 \to p_2}$ is True.

We will prove $p_2 \to p_3$ is True by a direct proof

$\boxed{\text{Case 2:}}$ Let p_2 be True. $\therefore \frac{a+b}{2} > a \Rightarrow \frac{b}{2} > a - \frac{a}{2} \Rightarrow \frac{b}{2} > \frac{a}{2} \Rightarrow \frac{b}{2} + \frac{b}{2} > \frac{a}{2} + \frac{b}{2}$

$\Rightarrow \boxed{b > \frac{a+b}{2}}$. $\therefore p_3$ is True. By defn. of conditional, $\boxed{p_2 \to p_3}$ is True.

We will prove $p_3 \to p_1$ is True by a direct proof

$\boxed{\text{Case 3:}}$ Let p_3 be True. $\therefore \frac{a+b}{2} < b \Rightarrow \frac{a}{2} < b - \frac{b}{2} \Rightarrow \frac{a}{2} < \frac{b}{2} \Rightarrow \boxed{a < b}$

$\therefore p_1$ is True. By defn. of conditional, $\boxed{p_3 \to p_1}$ is True.

$\boxed{\therefore \text{The 3 statements are equivalent}}$ □

33) Show that these statements about the integer x are equivalent:
(i) $3x+2$ is even (ii) $(x+5)$ is odd, (iii) x^2 is even.